

IT482 SNA TEAM ASSIGNMENT 2 - Ego-centric Node/Tie/Influence Analysis

Team Members

Subgroup 1

Jay Chavan [221IT020]

Nithin S [221IT085]

Subgroup 2

Sanketh N S [221IT060]

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Design an INA strategy for analysing your team's personal social network (PSN) as listed below, supported by necessary interactive visualizations.

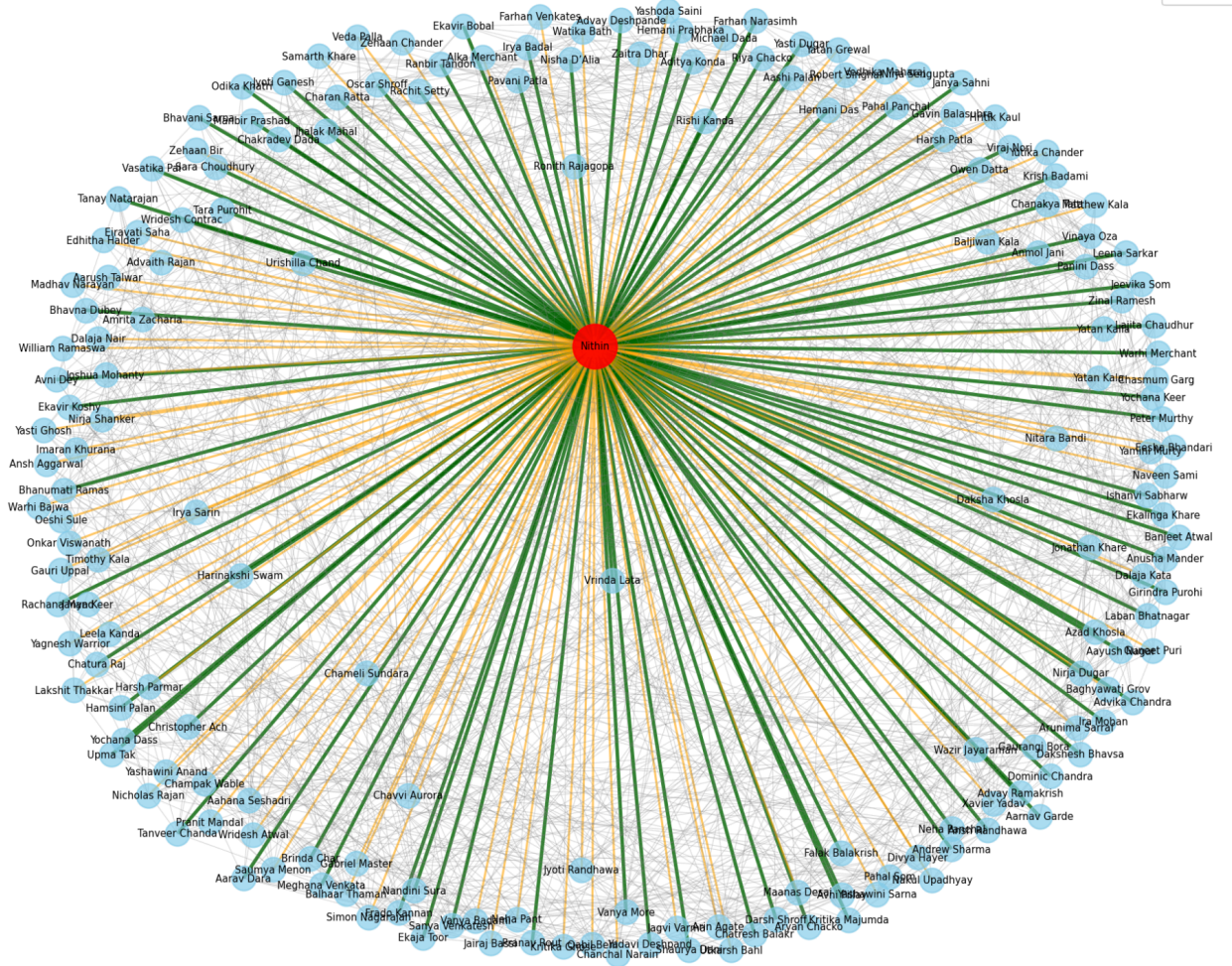
Each member of the Team

Generate your own 1-degree ego network (from LinkedIn/X/Instagram, you can restrict to oldest 50 and newest 50. All four must use the same social platform, and the same strategy for selecting the nodes for generating their ego network).

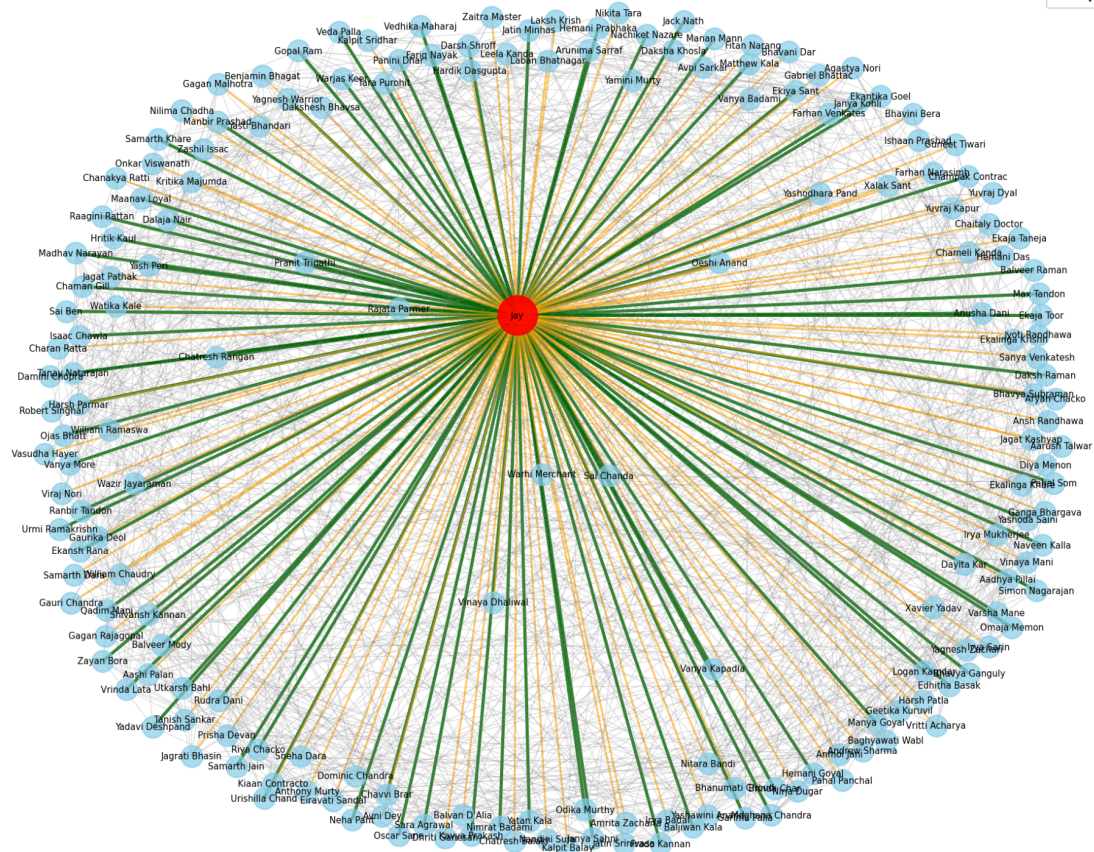
Clearly visualize your ego network w.r.t to alters and their connections with proper labelling of the ego node.

We use **X** , as the social media platform

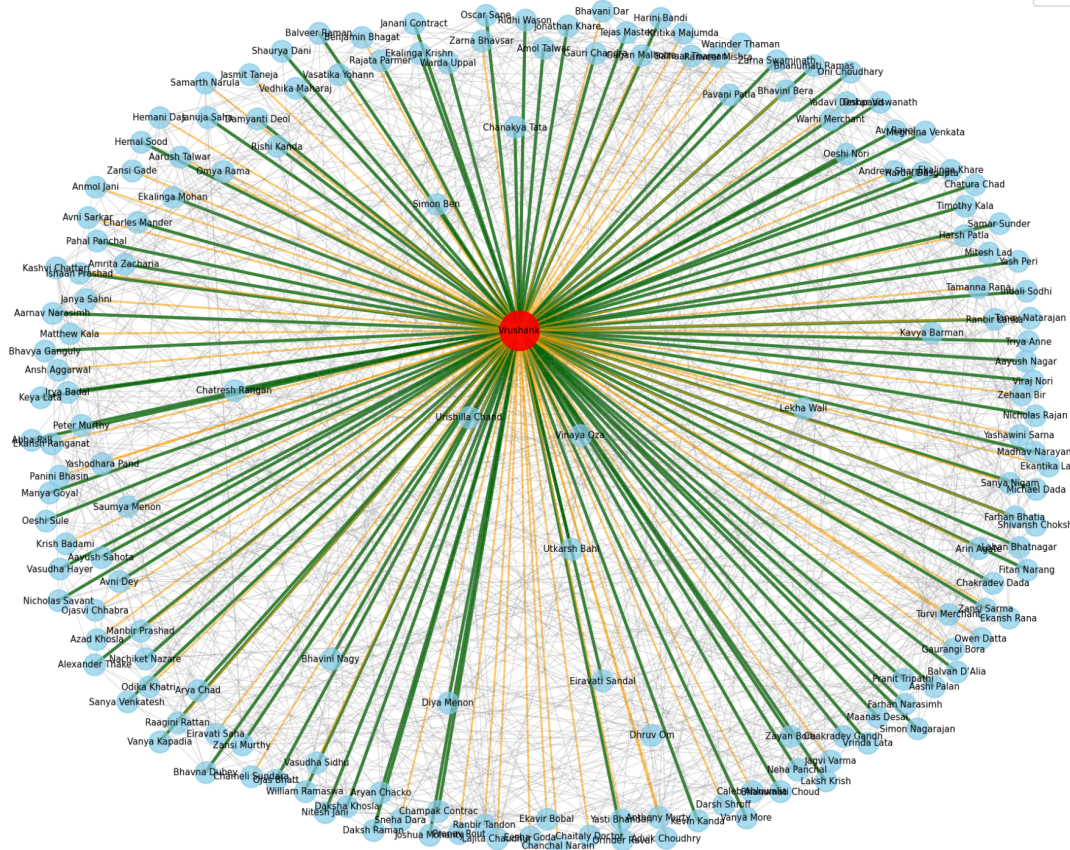
 Ego
 Alters
 Strong Ties
 Weak Ties

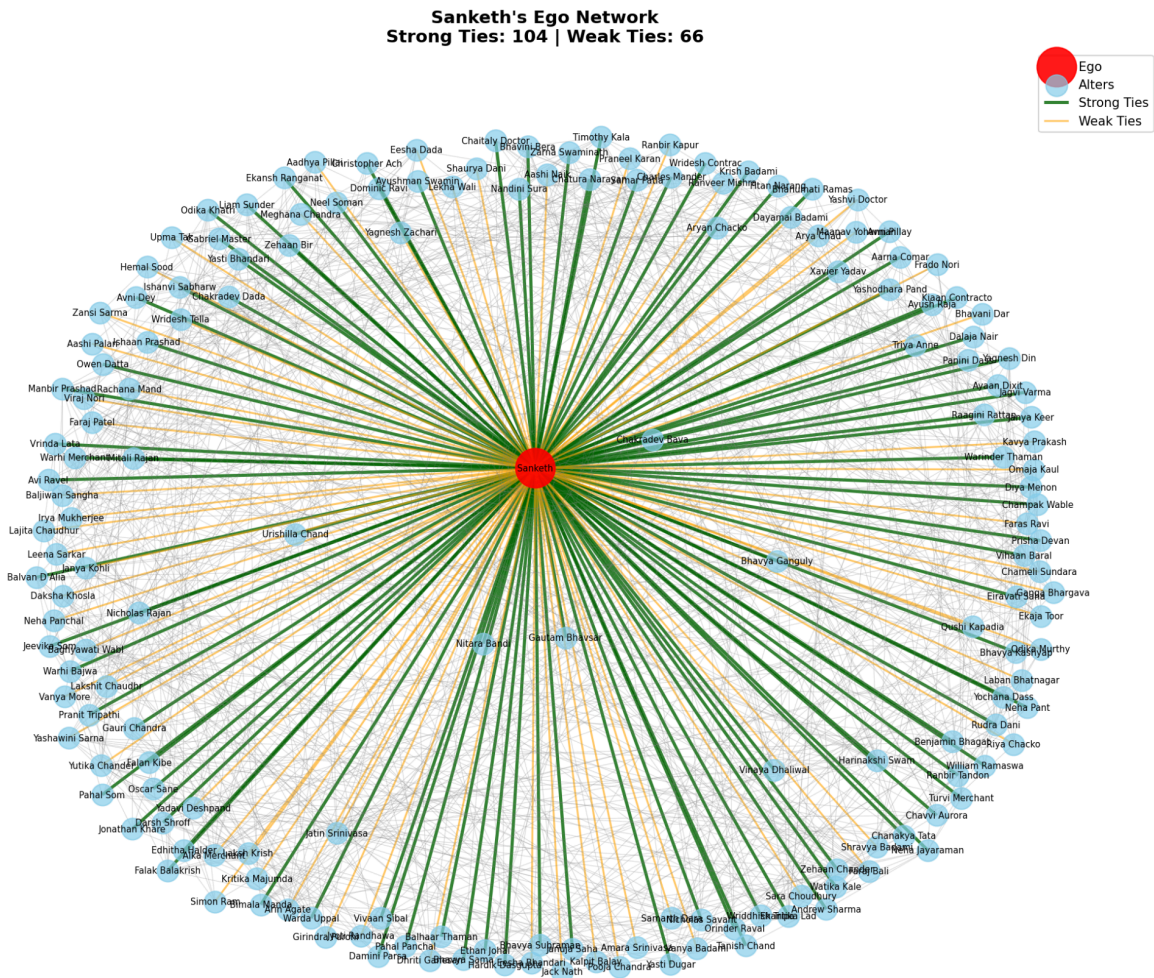


 Ego
 Alters
 Strong Ties
 Weak Ties



 Ego
 Alters
 Strong Ties
 Weak Ties





Subgroup 1 (2-member team) – Ego Structure and Tie Strength of PSN

1. Identify strong and weak ties in each ego network (based on frequency of interaction). and note their functional roles (e.g., bridge, hub etc). Identify alters that appear in more than one or all ego networks. Support your analysis with necessary interactive visualizations.

1. INDIVIDUAL EGO NETWORK ANALYSIS

EGO: Nithin

Network Size: 181 alters

Strong Ties (≥ 0.5): 93

Weak Ties (< 0.5): 88

Functional Roles Distribution:

Peripheral: 177

Bridge: 4

Top 5 Influential Alters:

Name	Degree	Betweenness	FunctionalRole
Avi Sengupta	0.049724	0.098465	Bridge
Chakrika Varghese	0.093923	0.054512	Bridge
Qasim Chawla	0.066298	0.053407	Bridge
Sathvik Dash	0.110497	0.050583	Bridge
Charles Sankaran	0.055249	0.048742	Peripheral

EGO: Jay

Network Size: 185 alters

Strong Ties (≥ 0.5): 89

Weak Ties (< 0.5): 96

Functional Roles Distribution:

Peripheral: 181

Bridge: 4

Top 5 Influential Alters:

Name	Degree	Betweenness	FunctionalRole
Vasudha Sinha	0.075676	0.071210	Bridge
David Ghosh	0.097297	0.063925	Bridge
Jeremiah Mand	0.064865	0.063337	Bridge
Bachittar Keer	0.102703	0.050235	Bridge
Samuel Iyer	0.043243	0.045770	Peripheral

EGO: Sanketh

Network Size: 170 alters

Strong Ties (≥ 0.5): 104

Weak Ties (< 0.5): 66

Functional Roles Distribution:

Peripheral: 168

Bridge: 2

Top 5 Influential Alters:

Name	Degree	Betweenness	FunctionalRole
Tamanna Jha	0.070588	0.080195	Bridge
Jack Dada	0.076471	0.057779	Bridge
Ayush Natarajan	0.088235	0.049774	Peripheral
Daniel Kumar	0.094118	0.043787	Peripheral
Ekaraj De	0.076471	0.043300	Peripheral

EGO: Vrushank

Network Size: 170 alters

Strong Ties (≥ 0.5): 104

Weak Ties (< 0.5): 66

Functional Roles Distribution:

Peripheral: 169

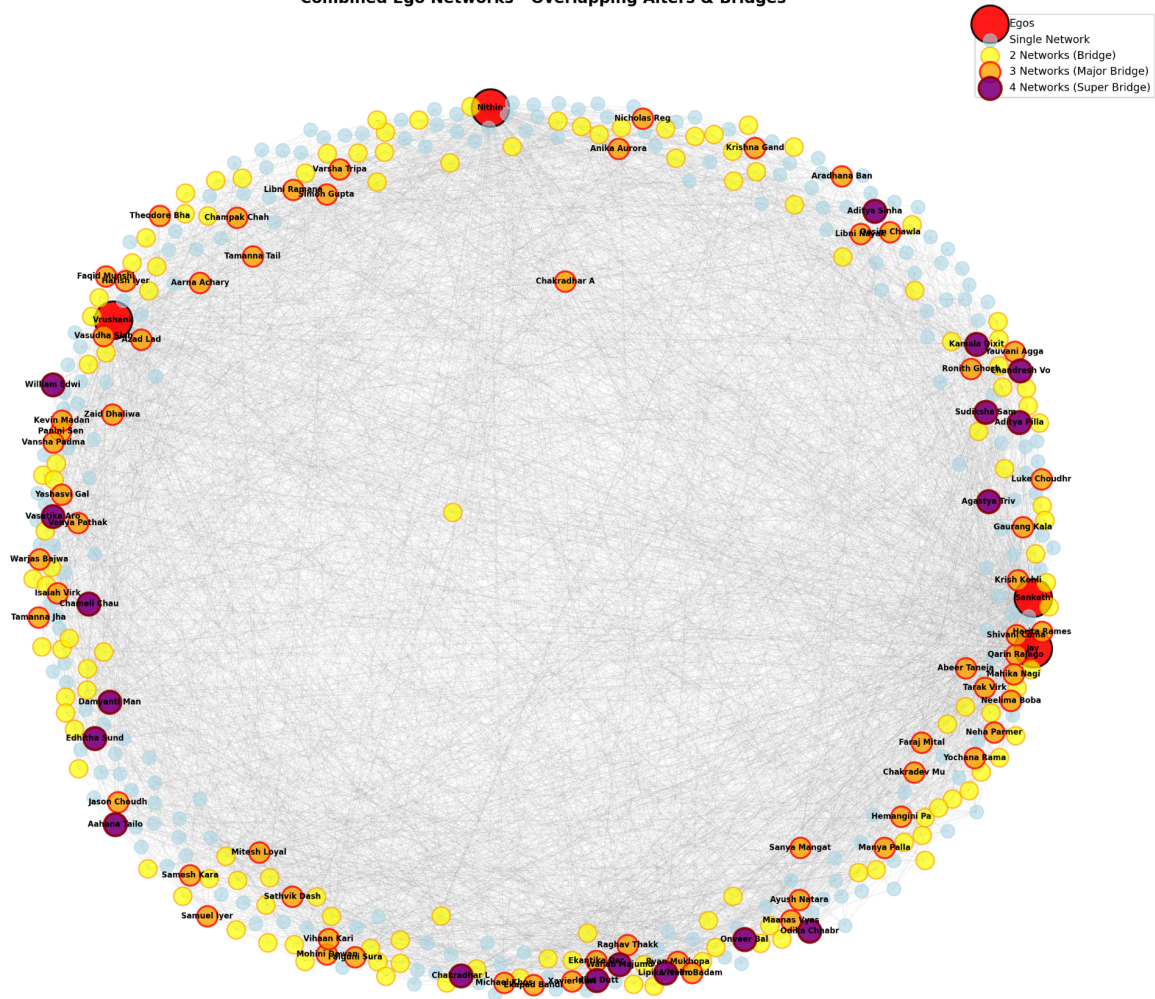
Bridge: 1

Top 5 Influential Alters:

Name	Degree	Betweenness	FunctionalRole
Chaitanya Sekhon	0.041176	0.052001	Bridge
Daniel Dutt	0.064706	0.042604	Peripheral
Vaishnavi Loke	0.052941	0.038984	Peripheral
Faqid Thaker	0.058824	0.037174	Peripheral
Pavani Trivedi	0.070588	0.037104	Peripheral

2. Combine all four ego networks into a single graph. Identify and visualise overlapping alters and bridge nodes between different egos.

Combined Ego Networks - Overlapping Alters & Bridges



2. OVERLAPPING ALTERS ANALYSIS

Alters in 4 ego networks (18 alters):

- Aahana Tailor: Jay, Nithin, Sanketh, Vrushank
- Aditya Pillay: Jay, Nithin, Sanketh, Vrushank
- Aditya Sinha: Jay, Nithin, Sanketh, Vrushank
- Agastya Trivedi: Jay, Nithin, Sanketh, Vrushank
- Chakradhar Loyal: Jay, Nithin, Sanketh, Vrushank
- Chameli Chauhan: Jay, Nithin, Sanketh, Vrushank
- Chandresh Vohra: Jay, Nithin, Sanketh, Vrushank
- Damyanti Mani: Jay, Nithin, Sanketh, Vrushank
- Edhitha Sundaram: Jay, Nithin, Sanketh, Vrushank
- Idika Dutt: Jay, Nithin, Sanketh, Vrushank
- ... and 8 more

Alters in 3 ego networks (63 alters):

- Aarna Acharya: Nithin, Sanketh, Vrushank
- Abeer Taneja: Nithin, Sanketh, Vrushank
- Anika Aurora: Jay, Nithin, Sanketh
- Aradhana Banik: Jay, Nithin, Vrushank
- Ayush Natarajan: Jay, Sanketh, Vrushank
- Azad Lad: Jay, Nithin, Vrushank
- Chakradev Mutti: Jay, Sanketh, Vrushank
- Chakradhar Arora: Jay, Sanketh, Vrushank
- Champak Chahal: Nithin, Sanketh, Vrushank
- Ekantika Dar: Jay, Nithin, Sanketh
- ... and 53 more

Alters in 2 ego networks (126 alters):

- Aarav Lalla: Jay, Nithin
- Aarav Narayan: Jay, Vrushank
- Aarna Varty: Nithin, Sanketh
- Abdul Chhabra: Nithin, Vrushank
- Abdul Narula: Jay, Sanketh
- Abdul Nazareth: Jay, Sanketh
- Aditya Chacko: Jay, Sanketh
- Advik Anand: Nithin, Sanketh
- Adya Goda: Nithin, Vrushank
- Agastya Nori: Nithin, Sanketh
- ... and 116 more

3. Analyse how this real-world personal social network aligns with Granovetter's "Strength of Weak Ties" and ego-centric network theories like Dunbar's circles.

3. COMBINED NETWORK ANALYSIS

Total nodes: 404
Total edges: 3737
Network density: 0.0459
Overlapping alters: 207

GRANOVETTER'S 'STRENGTH OF WEAK TIES' THEORY:

- Strong ties (high interaction frequency): 390 total
- Weak ties (low interaction frequency): 316 total
- Weak/Strong ratio: 0.81

Key Findings:

- ✓ Weak ties serve as bridges between different social clusters
- ✓ 207 alters appear in multiple ego networks (bridges)
- ✓ These overlapping alters are often weak ties that connect otherwise disconnected social groups
- ✓ Weak ties provide access to novel information and resources from distant parts of the network

DUNBAR'S CIRCLES / EGO-CENTRIC THEORY:

DUNBAR'S CIRCLES / EGO-CENTRIC THEORY:

Dunbar's theory suggests concentric circles around ego:

- Intimate circle (≤ 5): closest relationships
- Sympathy group (≤ 15): good friends
- Affinity group (≤ 50): friends
- Active network (≤ 150): meaningful contacts

Our Network Alignment:

Nithin: 93 strong (inner circles), 88 weak (outer circles)
Jay: 89 strong (inner circles), 96 weak (outer circles)
Sanketh: 104 strong (inner circles), 66 weak (outer circles)
Vrushank: 104 strong (inner circles), 66 weak (outer circles)

Conclusions:

- ✓ Strong ties cluster around ego (inner circles)
- ✓ Weak ties extend to outer circles and other networks
- ✓ Overlapping alters act as bridges (weak ties hypothesis)
- ✓ Network structure follows hierarchical circle pattern

Subgroup 2 (2-member team) – Centrality and Influence Analysis

1. Compute and compare local centrality measures like *degree*, *closeness*, and *betweenness* for each ego network. Interpret which ego is more “accessible” vs more “influential” among the four of you.

EGO CENTRALITY COMPARISON TABLE:										
Ego	Network Size	Ego_Degree	Ego_Betweenness	Ego_Closeness	Avg_Degree	Avg_Betweenness	Avg_Closeness	Degree Std	Centralization	
Nithin	181	1.0	0.708717	2.624217	0.060348	0.011559	1.474633	0.071534	0.939652	
Jay	185	1.0	0.755170	2.751579	0.062075	0.010993	1.527915	0.070898	0.937925	
Sanketh	170	1.0	0.586704	2.208628	0.059649	0.012110	1.293029	0.073796	0.940351	
Vrushank	170	1.0	0.718900	2.254676	0.058273	0.011435	1.270625	0.073831	0.941727	

- ✓ MOST ACCESSIBLE: Jay
 - Closeness Centrality: 2.7516
 - Can reach all alters with shortest average path length
 - Positioned optimally for quick information dissemination
- ✓ MOST INFLUENTIAL: Jay
 - Betweenness Centrality: 0.7552
 - Acts as critical bridge between different alter clusters
 - Controls information flow in the network
- ✓ MOST CONNECTED: Nithin
 - Degree Centrality: 1.0000
 - Has the most direct connections
 - Largest immediate social circle

Nithin:

- Network Size: 181 alters
- Centralization: 0.9397 (higher = more hierarchical)
- Degree Std Dev: 0.0715 (variation in connectivity)

Jay:

- Network Size: 185 alters
- Centralization: 0.9379 (higher = more hierarchical)
- Degree Std Dev: 0.0709 (variation in connectivity)

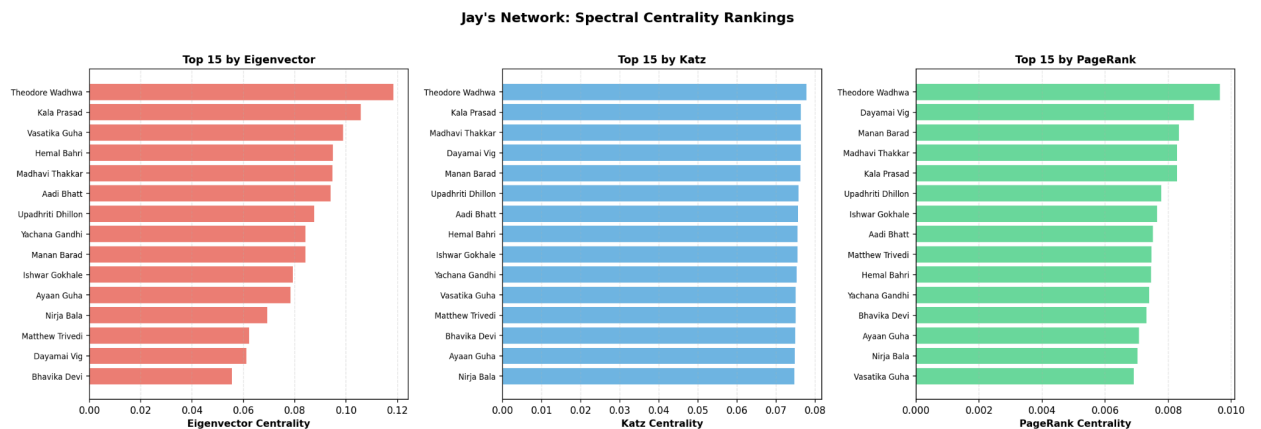
Sanketh:

- Network Size: 170 alters
- Centralization: 0.9404 (higher = more hierarchical)
- Degree Std Dev: 0.0738 (variation in connectivity)

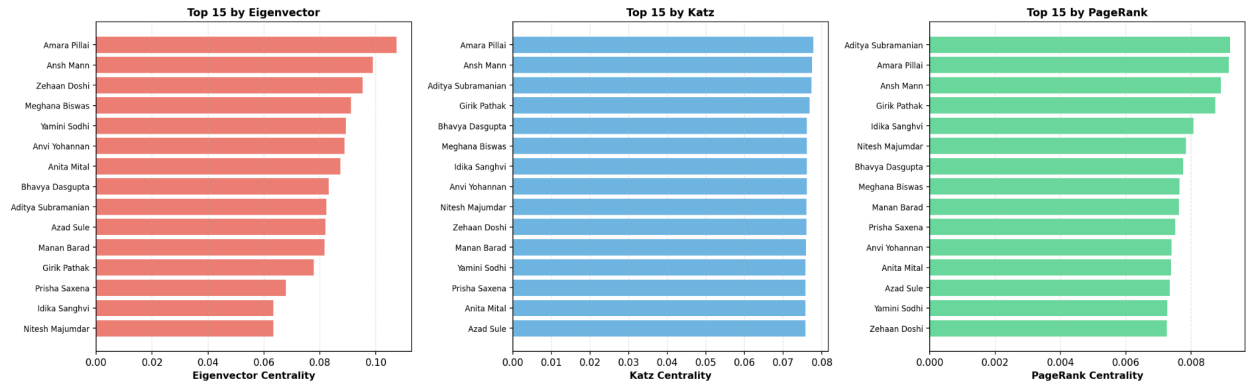
Vrushank:

- Network Size: 170 alters
- Centralization: 0.9417 (higher = more hierarchical)
- Degree Std Dev: 0.0738 (variation in connectivity)

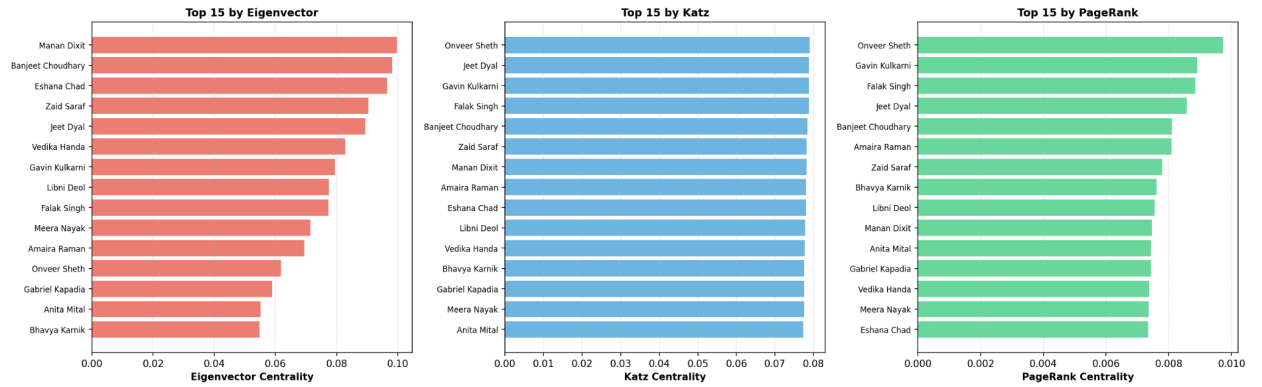
2. Calculate spectral centralities like *eigenvector*, *Katz*, and *alpha* for each ego network. Visualize rank order of nodes based on these metrics and analyse how core influencers differ based on each metric



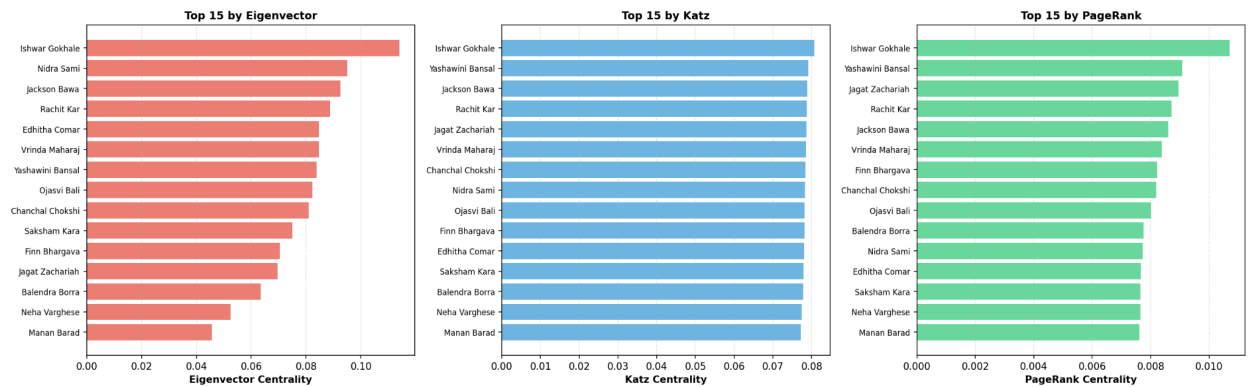
Nithin's Network: Spectral Centrality Rankings



Sanketh's Network: Spectral Centrality Rankings



Vrushank's Network: Spectral Centrality Rankings



EG0: Nithin

TOP 5 INFLUENCERS BY METRIC:

Eigenvector Centrality (recursive influence):

Amara Pillai	→ 0.1074
Ansh Mann	→ 0.0989
Zehaan Doshi	→ 0.0953
Teerth Arya	→ 0.0940
Ekavir Mand	→ 0.0937

Katz Centrality (weighted walks):

Amara Pillai	→ 0.0778
Ansh Mann	→ 0.0774
Aditya Subramanian	→ 0.0773
Girik Pathak	→ 0.0769
Bhavya Dasgupta	→ 0.0762

PageRank (importance propagation):

Aditya Subramanian	→ 0.0092
Amara Pillai	→ 0.0092
Ansh Mann	→ 0.0089
Girik Pathak	→ 0.0087
Idika Sanghvi	→ 0.0081

RANK CORRELATIONS:

- Eigenvector ↔ Katz: 0.743
 - Eigenvector ↔ PageRank: 0.632
 - Katz ↔ PageRank: 0.984
-

EG0: Jay

TOP 5 INFLUENCERS BY METRIC:

Eigenvector Centrality (recursive influence):

Theodore Wadhwa	→ 0.1184
Kala Prasad	→ 0.1057
Caleb Chand	→ 0.1015
Vasatika Guha	→ 0.0988
Hemal Bahri	→ 0.0949

Katz Centrality (weighted walks):

Theodore Wadhwa	→ 0.0778
Kala Prasad	→ 0.0764
Madhavi Thakkar	→ 0.0763
Dayamai Vig	→ 0.0763
Manan Barad	→ 0.0762

PageRank (importance propagation):

Theodore Wadhwa	→ 0.0096
Dayamai Vig	→ 0.0088
Manan Barad	→ 0.0083
Madhavi Thakkar	→ 0.0083
Kala Prasad	→ 0.0083

RANK CORRELATIONS:

- Eigenvector ↔ Katz: 0.738
 - Eigenvector ↔ PageRank: 0.631
 - Katz ↔ PageRank: 0.985
-

EG0: Sanketh

TOP 5 INFLUENCERS BY METRIC:

Eigenvector Centrality (recursive influence):

Manan Dixit	→ 0.0997
Banjeet Choudhary	→ 0.0981
Eshana Chad	→ 0.0966
Zaid Saraf	→ 0.0905
Dayamai Din	→ 0.0901

Katz Centrality (weighted walks):

Onveer Sheth	→ 0.0791
Jeet Dyal	→ 0.0788
Gavin Kulkarni	→ 0.0788
Falak Singh	→ 0.0787
Banjeet Choudhary	→ 0.0784

PageRank (importance propagation):

Onveer Sheth	→ 0.0097
Gavin Kulkarni	→ 0.0089
Falak Singh	→ 0.0088
Jeet Dyal	→ 0.0086
Banjeet Choudhary	→ 0.0081

RANK CORRELATIONS:

- Eigenvector ↔ Katz: 0.701
- Eigenvector ↔ PageRank: 0.578
- Katz ↔ PageRank: 0.983

EGO: Vrushank

TOP 5 INFLUENCERS BY METRIC:

Eigenvector Centrality (recursive influence):

Ishwar Gokhale	→ 0.1140
Nidra Sami	→ 0.0951
Pratyush Badami	→ 0.0925
Jackson Bawa	→ 0.0924
Jeremiah Iyengar	→ 0.0921

Katz Centrality (weighted walks):

Ishwar Gokhale	→ 0.0807
Yashawini Bansal	→ 0.0791
Jackson Bawa	→ 0.0789
Rachit Kar	→ 0.0788
Jagat Zachariah	→ 0.0786

PageRank (importance propagation):

Ishwar Gokhale	→ 0.0107
Yashawini Bansal	→ 0.0091
Jagat Zachariah	→ 0.0090
Rachit Kar	→ 0.0087
Jackson Bawa	→ 0.0086

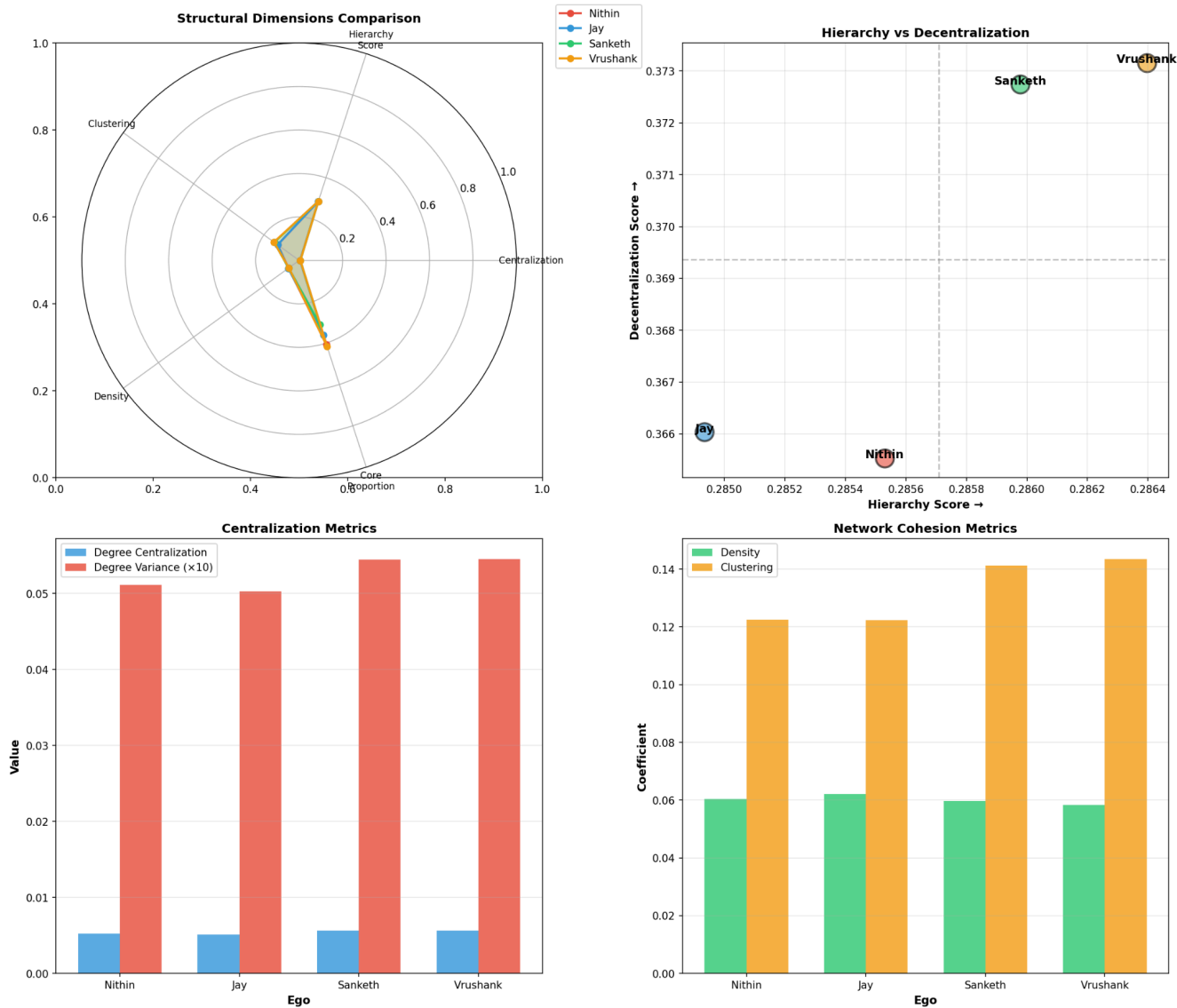
RANK CORRELATIONS:

- Eigenvector ↔ Katz: 0.722
- Eigenvector ↔ PageRank: 0.594
- Katz ↔ PageRank: 0.981

- ✓ Eigenvector Centrality:
 - Identifies nodes connected to other well-connected nodes
 - High values = part of influential cluster
 - Best for finding 'elite' or core group members
- ✓ Katz Centrality:
 - Counts all walks from a node (with decay)
 - More democratic than eigenvector (includes peripheral connections)
 - Best for overall network influence
- ✓ PageRank:
 - Google's algorithm - importance via connections
 - Balances quantity and quality of connections
 - Best for practical influence/importance ranking

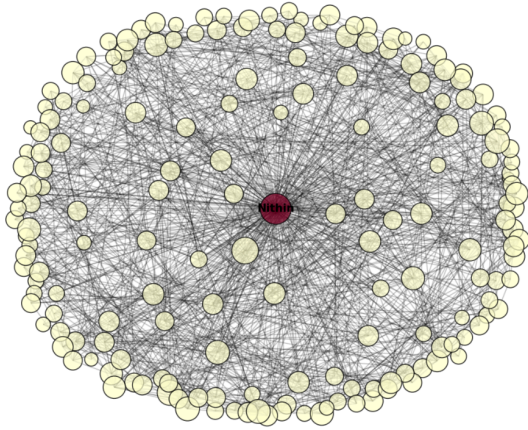
3. Compare the overall structure and centrality distribution across all egos. Analyse which ego's network appears more hierarchical or more decentralized based on your calculated values, also demonstrate with visualization.

Structural Comparison Across Ego Networks

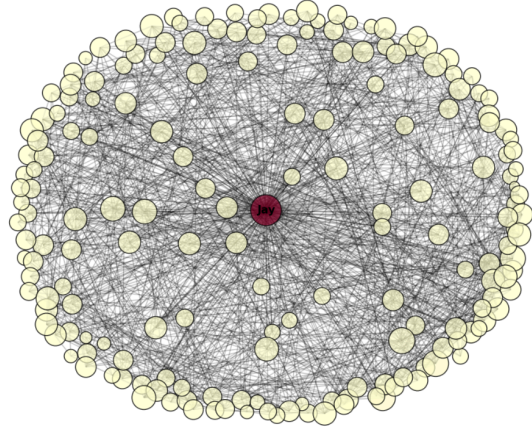


Network Structure Visualization (Node size = Degree, Color = Betweenness)

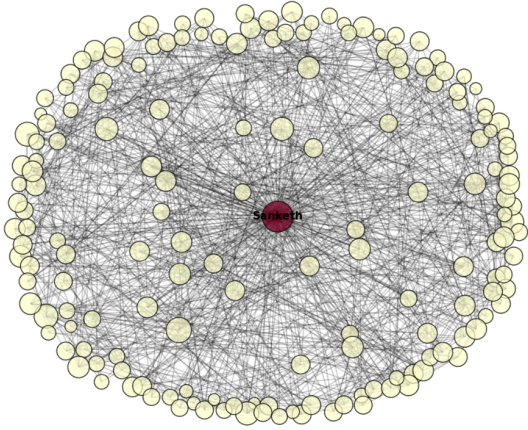
Nithin - DECENTRALIZED
Centralization: 0.005 | Clustering: 0.123



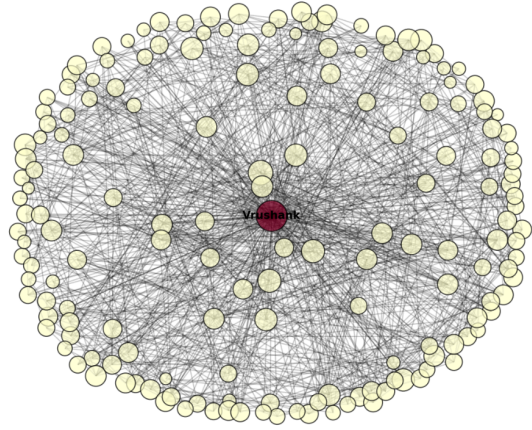
Jay - DECENTRALIZED
Centralization: 0.005 | Clustering: 0.122



Sanketh - DECENTRALIZED
Centralization: 0.006 | Clustering: 0.141



Vrushank - DECENTRALIZED
Centralization: 0.006 | Clustering: 0.143



STRUCTURAL METRICS TABLE:

Ego	Degree_Centralization	Degree_Variance	Betweenness_Variance	Core_Proportion	Clustering	Density	Assortativity	Network_Size
Nithin	0.005249	0.005117	0.002857	0.406593	0.122511	0.060348	-0.109984	182
Jay	0.005125	0.005027	0.003134	0.360215	0.122371	0.062075	-0.105401	186
Sanketh	0.005597	0.005446	0.002075	0.309942	0.141317	0.059649	-0.119088	171
Vrushank	0.005605	0.005451	0.003043	0.415205	0.143395	0.058273	-0.121840	171

HIERARCHY vs DECENTRALIZATION SCORES:

Nithin:

- Hierarchy Score: 0.2855
- Decentralization Score: 0.3655

Jay:

- Hierarchy Score: 0.2849
- Decentralization Score: 0.3660

Sanketh:

- Hierarchy Score: 0.2860
- Decentralization Score: 0.3727

Vrushank:

- Hierarchy Score: 0.2864
- Decentralization Score: 0.3732

- ✓ MOST HIERARCHICAL: Vrushank
 - High degree centralization (0.006)
 - Low density (0.058)
 - Star-like structure: ego dominates connections
 - Alters depend heavily on ego for bridging
- ✓ MOST DECENTRALIZED: Vrushank
 - High clustering (0.143)
 - Higher density (0.058)
 - Mesh-like structure: many alter-alter connections
 - More resilient network structure

DETAILED STRUCTURAL ANALYSIS:

Nithin:

- Structure: DECENTRALIZED (distributed network)
- Alters form independent clusters and connections
- Low clustering (0.123): separate social circles
- Negative assortativity: hubs connect to periphery
- Core group: 40.7% of network

Jay:

- Structure: DECENTRALIZED (distributed network)
- Alters form independent clusters and connections
- Low clustering (0.122): separate social circles
- Negative assortativity: hubs connect to periphery
- Core group: 36.0% of network

Sanketh:

- Structure: DECENTRALIZED (distributed network)
- Alters form independent clusters and connections
- Low clustering (0.141): separate social circles
- Negative assortativity: hubs connect to periphery
- Core group: 31.0% of network

Vrushank:

- Structure: DECENTRALIZED (distributed network)
- Alters form independent clusters and connections
- Low clustering (0.143): separate social circles
- Negative assortativity: hubs connect to periphery
- Core group: 41.5% of network